

A Novel Automatic Framework for Speaker Drift Detection in Synthesized Speech

Jia-Hong Huang^{1,4}, Seulgi Kim², Yi Chieh Liu^{2,4}, Yixian Shen¹, Hongyi Zhu¹, Prayag Tiwari³, Stevan Rudinac¹, Evangelos Kanoulas¹

¹University of Amsterdam, ²Georgia Institute of Technology, ³Halmstad University, ⁴Amazon AGI

1. Research Background & Motivation

Background and Motivation

- Diffusion-based TTS achieves high naturalness and expressiveness
- Widely used in long-form speech and interactive systems
- Speaker consistency is critical for user experience
- Existing evaluation mainly focuses on audio quality

Gap

- Speaker identity consistency is underexplored

2. Problem Definition

Problem Definition: Speaker Drift

- Speaker drift: gradual identity shift within a single utterance
- Occurs even with fixed speaker conditioning
- Different from:
 - speaker diarization (abrupt changes)
 - speaker verification (cross-utterance)

Challenges

- Subtle and continuous
- Hard to quantify automatically
- No standard dataset or evaluation

Task	Goal	Key Characteristics
Speaker Change Detection	Identify speaker boundaries in natural multi-speaker speech streams	Assumes abrupt, discrete speaker transitions; aims at accurate change-point detection
Speaker Verification or Identification	Confirm or classify speaker identity using labeled reference data	Requires ground-truth speaker labels; evaluates similarity across separate utterances
Speaker Diarization	Segment and cluster speech by speaker in multi-speaker audio	Operates on known or unknown speakers; assumes clear speaker turns; insensitive to subtle intra-speaker drift
Voice Cloning Consistency Evaluation	Evaluate preservation of speaker identity in synthetic speech	Primarily relies on subjective human assessments or limited ABX testing; lacks automated, fine-grained detection methods
Out-of-Distribution Detection (Speaker Embeddings)	Detect anomalous speaker embeddings outside known distribution	Depends on large reference datasets; unsuitable for fine-grained temporal consistency within utterances
Voice Style Transfer Consistency	Assess preservation of non-identity attributes such as prosody, emotion, or accent	Focuses on style or affective features; does not explicitly target speaker identity stability
Speaker Drift Detection (Ours)	Detect subtle, gradual variations in speaker identity within a single utterance, particularly in synthetic speech from TTS systems	Reference-free; focuses on temporal embedding consistency within utterances; addresses identity stability challenges unique to TTS

Table 1: Comparison of speaker drift detection with related tasks in speaker and TTS research.

3. Proposed Framework

Overview

- Segment synthesized speech into consecutive parts
- Extract speaker embeddings
- Compute cosine similarity between segments
- Use LLM for drift classification

Pipeline

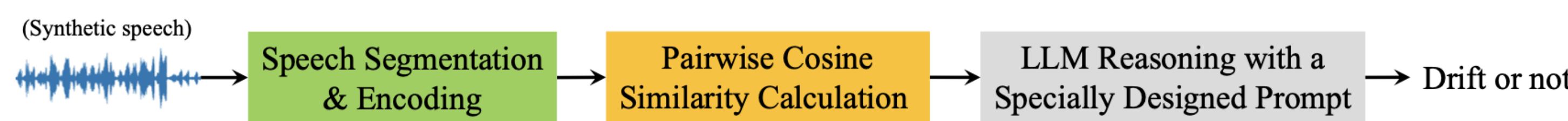


Figure 1: Cross-entropy loss and multi-class accuracy as a function of the training epoch for the classical and quantum LSTM Part-of- Speech taggers.

Key Design

- Use similarity instead of raw embeddings
- Structured prompt for LLM
- Reference-free evaluation

Instruction template:

You are given a list of N pairwise similarity scores derived from three consecutive segments of each utterance. Each score reflects the similarity between adjacent segments, computed using {your similarity metric}, with embeddings obtained from {your audio encoder}. Your task is to assess whether each utterance exhibits speaker drift based on the provided similarity scores. For instance, consider the example: (0.9963, 0.9872). Based on this, determine whether a speaker identity shift has likely occurred. Provide a binary decision, same (no drift) or different (drift), and briefly explain your reasoning. Evaluate each case independently and report your decisions with corresponding justifications.

4. Key Insight & Theory

Insight

- Same speaker $\rightarrow$ embeddings cluster
- Drift $\rightarrow$ similarity decreases

Theory

- Cosine similarity provides a reliable signal
- Threshold-based classifier has bounded error

Proposition 1 (Embedding Smoothness & Drift Separability). Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^d$ be embeddings from three contiguous speech segments. Define pairwise cosine similarities $\text{sim}_{i,j} = \cos(\mathbf{e}_i, \mathbf{e}_j)$. Suppose: $\max(\text{sim}_{1,2}, \text{sim}_{2,3}) < \tau$ for some threshold $\tau \in (0, 1)$. Then any classifier f that is Lipschitz-continuous over the similarity space can separate drift from non-drift samples with bounded error: $\mathbb{P}(f(\text{sim}_{1,2}, \text{sim}_{2,3}) \neq y) \leq \epsilon(\tau)$, where $\epsilon(\tau) \rightarrow 0$ as $\tau \rightarrow 0$, assuming embeddings vary smoothly for the same speaker and show sufficient separation across speakers.

Theorem 1 (Embedding Separation Bound for Speaker Drift Detection). Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{S}^{d-1}$ denote unit-norm embeddings corresponding to three contiguous speech segments. We assume that for non-drift (i.e., same-speaker) samples, the expected pairwise cosine similarity satisfies $\mathbb{E}[\cos(\mathbf{e}_i, \mathbf{e}_j)] \geq \mu_0$ with variance bounded by σ^2 . In contrast, for drift (i.e., different-speaker) samples, the expected similarity is lower, satisfying $\mathbb{E}[\cos(\mathbf{e}_i, \mathbf{e}_j)] \leq \mu' < \mu_0$, with the same variance σ^2 . Define the classifier:

$$f(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) = \begin{cases} 1 & \text{if } \min(\cos(\mathbf{e}_1, \mathbf{e}_2), \cos(\mathbf{e}_2, \mathbf{e}_3)) < \tau, \\ 0 & \text{otherwise,} \end{cases}$$

for any threshold $\tau \in (\mu', \mu_0)$. Then, the misclassification probability is bounded by:

$$\mathbb{P}(f(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \neq y) \leq 4 \exp\left(-\frac{\Delta^2}{2\sigma^2}\right), \quad \text{where} \\ \Delta = \min(\mu_0 - \tau, \tau - \mu').$$

👉 This result provides a concrete error bound, showing that the classifier’s performance improves exponentially as the separation margin (Δ) grows.

5. Experiments

Dataset

- Synthetic benchmark with human verification
- Includes:
 - non-drift
 - abrupt drift
 - smooth drift
 - hard negatives

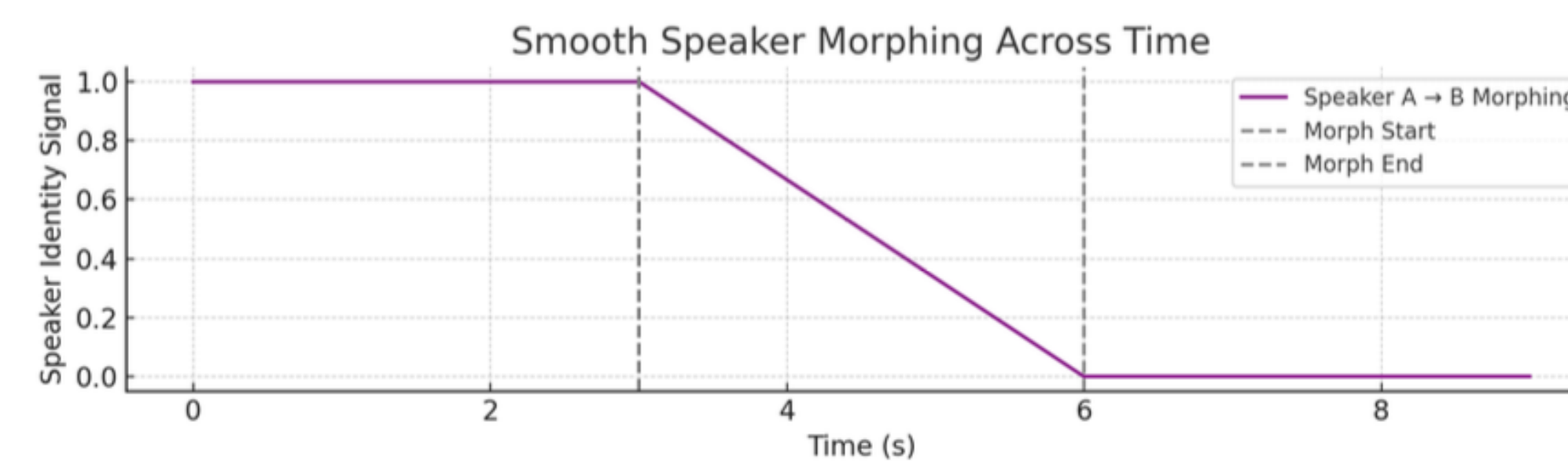


Figure 2: Smooth speaker morphing across an utterance. The morphing region (3s–6s) involves audio-level cross-fading from Speaker A to Speaker B.

Results

- LLM + cosine similarity outperforms baselines
- Better than:
 - fixed threshold
 - PCA-based embedding input

Method	Input Type	Accuracy	F1 Score
GPT-4o	PCA Embeddings (8)	50.3%	66.7%
GPT-4o	PCA Embeddings (16)	73.4%	73.6%
GPT-4o	Cosine Scores	89.5%	90.7%
Gemini-Pro-2.5	PCA Embeddings (8)	50.8%	58.3%
Gemini-Pro-2.5	PCA Embeddings (16)	52.3%	59.8%
Gemini-Pro-2.5	Cosine Scores	79.7%	82.9%
Claude-4	PCA Embeddings (8)	63.7%	69.3%
Claude-4	PCA Embeddings (16)	67.5%	73.6%
Claude-4	Cosine Scores	83.4%	88.2%
Qwen-3	PCA Embeddings (8)	65.3%	68.0%
Qwen-3	PCA Embeddings (16)	69.1%	72.4%
Qwen-3	Cosine Scores	69.5%	72.7%
DeepSeek-R1	PCA Embeddings (8)	60.9%	71.2%
DeepSeek-R1	PCA Embeddings (16)	63.4%	74.4%
DeepSeek-R1	Cosine Scores	78.8%	80.6%

Table 3: Ablation study of the proposed LLM-driven speaker drift detection framework, evaluating the impact of different LLM backbones and input feature formats.

Embedding Type	GPT-4o [18]	Gemini-Pro-2.5 [19]	Claude-4 [20]	Qwen-3 [22]	DeepSeek-R1 [21]	PCA-based Baseline	Fixed-threshold Baseline
Win2Vec2 [23]	90.70%	82.90%	88.20%	72.70%	80.00%	71.30%	61.70%
MFCC [24]	39.00%	38.64%	76.00%	71.40%	80.00%	65.60%	57.40%
Whisper [25]	89.41%	80.00%	84.38%	72.70%	80.00%	90.60%	61.30%

Table 2: Performance comparison with baselines and ablation studies using different audio embedding methods, evaluated using F1 score and pairwise cosine similarity.

6. Conclusion & Future Work

Conclusion

- First automatic framework for speaker drift detection
- Combines embedding geometry + LLM reasoning
- Enables scalable evaluation for TTS

Future Work

- Multilingual and cross-lingual settings
- Fine-tuned LLMs for better sensitivity
- Real-world deployment in TTS pipelines